

v5 Binding Model — Downstream Output Format & Partition-Function Design

Design discussion notes

2026-07-15

Scope

Working notes from a design discussion about how the v5 HSMM binding classifier (`classify_single_molecule_binding_v5_hsmm.py`) output should feed downstream analyses, and how to reparametrize the thermodynamic (partition-function) model to consume it. Covers: the v5 output schema and its compatibility with old code, how to make single recurrent footprints callable, a plan for a v5-native aggregator, the nucleosome meta-plot denominator, the role of UNID footprints, and — the centerpiece — an Ising/MaxEnt reformulation of the partition function.

1. v5 output schema

Two files per (sample, amplicon).

Main per-molecule file (`{sample}.{amplicon}.single_molecule_classification.txt`, indexed by `read_id`):

- `tfbs_1 ... tfbs_K` — boolean, one per motif/TetO site
- `n_tf, n_nuc, n_unid` — counts
- `nucs` — semicolon-joined `start:end:dyad` intervals
- `unids` — semicolon-joined `start:end` intervals
- `log_likelihood` — total Viterbi path score

Segments file (`.segments.txt`, tidy long — one row per segment): `read_id, seg_index, type, start, end, dyad_or_motif, seg_loglik`.

This is deliberately **not** coerced back to the old bin-based schema. There is no `idx` column and no companion `valid_states_table.txt`, because the HSMM does not enumerate a global microstate space.

Old schema (v2 / v2_carosversion), for contrast: `tfbs_{k}`; ragged wide `nuc{N}_present/start/end`; and `idx` into `valid_states_table.txt`.

2. Compatibility with existing downstream code

Analysis	Works as-is?	Why
Binding histogram, <code>n_bound</code> , <code>tf_bound</code>	Yes	Both formats share <code>tfbs_{k}</code> ; <code>aggregate_binding_model.py</code> uses only <code>df.filter(like='tfbs_').sum(axis=1)</code> . None of v5's extra columns contain the substring <code>tfbs_</code> .
Peripheral accessibility / remodeling	Yes	Computed from the matrix, not the classification — classifier-independent.
Nucleosome occupancy	Adapt	Old <code>nuc{N}_*</code> columns gone; use the <code>nucs</code> string or <code>.segments.txt</code> .
Partition function fit	No	<code>fit_partition_function_model_v3.py</code> needs <code>valid_states_table.txt</code> + the <code>idx</code> join; v5 produces neither by design.
UNID footprints	New	No equivalent in the old format.

Decision: **do not** coerce v5 back to the old wide schema. The tidy `.segments.txt` is a strictly better substrate; write new, non-legacy analyses against it.

3. Making a single recurrent footprint callable

Why two adjacent ATF1 sites get called UNID but a single site does not. To open a footprint segment, Viterbi must accumulate enough protection evidence to pay the state's **entry cost**. With the v5 emission parameters, one protected GpC is worth roughly

$$\log \frac{P(\text{protected} \mid \text{UNID/TF})}{P(\text{protected} \mid \text{OPEN})} \approx \log \frac{0.95 \cdot 0.9 + 0.15 \cdot 0.1}{0.95 \cdot 0.05 + 0.15 \cdot 0.95} = \log \frac{0.87}{0.19} \approx +1.5.$$

The entry costs are `start_cost_unid` = 6.0 and `start_cost_tf` = 1.0. So:

- **Two adjacent sites** (about 3–4 protected GpCs, $\approx +4.5$ to $+6.0$) clear the UNID hurdle of 6.0.
- **A single site** (about 1–2 GpCs, $\approx +1.5$ to $+3.0$) does not — it stays OPEN.

Two ways to fix, both on the same “position-specific prior” lever:

1. **Discovery pass** (`--discovered_footprints_file`): lowers the UNID entry cost *at that locus* from 6.0 to ~ 2 , so a single site's evidence now clears it. Automated, no naming.
2. **Name it as a motif $\Rightarrow$ it calls as TF, not UNID** (preferred for a site you care about): entry cost drops to 1.0, so **one protected GpC (+1.5) already clears it**; and its duration

is the exact motif window rather than being bounded by `unid_min = 15 bp` (a single footprint narrower than 15 bp cannot form a legal UNID segment at all, but can be a TF of any width).

The “name everything eventually” endgame and the “make single sites callable” need are therefore the *same* lever.

4. A v5-native aggregator

New script (operating on the main file + `.segments.txt`), tidy at all three levels, losing no columns. Proposed per-molecule superset:

- **Keys:** `read_id`, `sample`, `amplicon`, `background`, `n_tfbs`
- **Binding (verbatim):** `tfbs_1...K`, `n_tf`, `tf_bound`
- **Nucleosome:** `n_nuc`, `nuc_bases_covered`, `frac_nuc`, `raw_nucs`
- **UNID:** `n_unid`, `unid_bases_covered`, `frac_unid`, `raw_unids`
- **Open:** `open_bases`, `frac_open` (sanity: fractions sum to 1)
- **Any footprint (the “is there a footprint here” query):** `footprint_bases_covered` = `NUC ∪ TF ∪ UNID`, `frac_footprint`
- **Kept from old:** `peripheral_accessibility` (from matrix), `log_likelihood`

Per-amplicon: means + count histograms of `n_tf/n_nuc/n_unid` (reuse the `n:freq` histogram-string trick). Per-sample: as today.

Decision to make explicit: `frac_*` needs a per-molecule denominator. Use decoded span (first-to-last GpC), not raw amplicon length, so molecules with different coverage compare fairly.

5. Nucleosome meta-plot denominator

Because Viterbi assigns a definite state to **every base of every molecule** (it fills through GpC gaps), nucleosome occupancy is *defined everywhere* — no -1/no-info denominator correction like on the raw accessibility matrix. So, per base *b*:

$$\text{occupancy}(b) = \frac{\#\{\text{molecules with a NUC segment spanning } b\}}{\#\{\text{total molecules}\}}.$$

The denominator comes straight from the segments file: every molecule has a full path, so every `read_id` appears (with its OPEN/TF segments even if it has zero NUCs). One footnote: a molecule with no GpC coverage in a region still decodes (toward OPEN) and counts as “not nucleosome,” slightly deflating occupancy there — mask per-molecule uninformative regions only if it matters.

6. Role of UNID footprints

Endgame: name recurrent footprints (like TetOs — “this NRF, this ATF”) so each becomes a named entry in the TF file. But UNID is **not** made obsolete:

- It is the **residual/discovery channel** — “recurrent protection I have not accounted for yet.” Deleting it forces the model to explain unexpected protection as OPEN (missed) or NUC (miscalled). It is how you find the next thing to name.
- The **“is there a footprint here” query** is not a UNID query — it is the per-bp occupancy of $\text{TF} \cup \text{UNID}$ ($\cup \text{NUC}$ if nucleosomes count). Named sites tell you *which* footprint; the union track tells you *whether* one is there. You want both, permanently.

So: build bulk/union tracks now, name sites incrementally into TF, and let UNID asymptote toward “residual” rather than “primary output.” It never has to reach zero to be done.

7. Partition function, Ising/MaxEnt style

7.1 The reframe

The literal microstate-enumeration approach *is* an Ising/MaxEnt model — just parametrized by enumeration. A microstate σ (which TetOs are TF-bound, where nucleosomes sit) is assigned an energy that is **linear in a set of interaction features** $\phi_k(\sigma)$:

$$E(\sigma; \theta) = \sum_k \theta_k \phi_k(\sigma),$$

with, for example,

$$\begin{aligned} \phi_{\text{tf},i}(\sigma) &= \mathbb{K}[\text{site } i \text{ TF-bound}] && (\text{per-TF energy / field } h_i) \\ \phi_{\text{coop},i}(\sigma) &= \mathbb{K}[\text{sites } i, i+1 \text{ both bound}] && (\text{cooperativity } J) \\ \phi_{\text{nuc}}(\sigma) &= \#\{\text{nucleosomes}\} \text{ or a region indicator} && (\text{per-nuc energy}) \\ \phi_{\text{tf-nuc}}(\sigma) &= \mathbb{K}[\text{TF and nuc overlap}] && (\text{TF-nucleosome antagonism}). \end{aligned}$$

The probability is exactly Boltzmann:

$$P(\sigma \mid \theta) = \frac{1}{Z(\theta)} e^{-E(\sigma; \theta)}, \quad Z(\theta) = \sum_{\sigma} e^{-E(\sigma; \theta)}.$$

This is identical to “energy per TF, energy per nuc, optional cooperative term” — a Gibbs distribution, i.e. the same object as a MaxEnt model and a generalized Ising model. Fields h_i are single-site energies; couplings J_{ij} are cooperativities.

7.2 The theorem that removes enumeration from *fitting*

The log-likelihood of observed molecules $\sigma_1, \dots, \sigma_N$ is

$$\ell(\theta) = -N \sum_k \theta_k \langle \phi_k \rangle_{\text{data}} - N \log Z(\theta), \quad \langle \phi_k \rangle_{\text{data}} = \frac{1}{N} \sum_m \phi_k(\sigma_m).$$

Using $\partial \log Z / \partial \theta_k = -\langle \phi_k \rangle_{\text{model}}$, the gradient is

$$\frac{\partial \ell}{\partial \theta_k} = N(\langle \phi_k \rangle_{\text{model}} - \langle \phi_k \rangle_{\text{data}}),$$

so the MLE is **moment matching**:

$$\langle \phi_k \rangle_{\text{model}}(\theta) = \langle \phi_k \rangle_{\text{data}}$$

Three consequences:

1. **The data enters only through feature averages** $\langle \phi_k \rangle_{\text{data}}$ — per-site occupancy, adjacent co-occupancy, mean nucleosome count — read directly off the v5 per-molecule calls. No microstate assignment, no `idx`.
2. **The fit is convex.** $\log Z$ is convex (log-sum-exp), so ℓ is concave in θ : unique optimum, gradient ascent suffices. The Hessian is $-N \text{Cov}_{\text{model}}(\phi)$, giving parameter error bars for free.
3. **No multiplicity factors.** $P(\sigma)$ is per-configuration; no $\binom{N}{n}$ degeneracy bookkeeping. The “SMF measures microstates directly” intuition is preserved exactly.

7.3 Where configurations enter — Z — and why 1D structure is cheap

Enumeration would only have been needed to compute Z and $\langle \phi_k \rangle_{\text{model}}$. The system is one-dimensional, and 1D lattice models have exact, cheap partition functions:

- **TetO array $\rightarrow$ transfer matrix.** With $E = \sum_i h_i n_i + \sum_i J n_i n_{i+1}$, Z is a product of 2×2 matrices — $O(L)$, exact, for any number of sites. Marginals $\langle n_i \rangle$ and co-occupancies $\langle n_i n_{i+1} \rangle$ (i.e. all $\langle \phi \rangle_{\text{model}}$) come from forward-backward on the chain. The combinatorial sum is done by the matrix product, correctly weighted, with no multiplicity bookkeeping.
- **Nucleosomes $\rightarrow$ 1D lattice gas of hard rods.** Extended excluded-volume objects on a 1D template — the same math as thermodynamic nucleosome-positioning models. Z is a forward DP over positions with a fugacity per nucleosome and hard-core exclusion. Exact, $O(\text{length})$.
- Brute enumeration is still permitted when a model is small; the point is Z is now a decoupled computational detail, not a prerequisite for touching the data.

You keep the literal partition function — still $\sum_{\sigma} e^{-E}$ — computed with a transfer matrix / DP instead of materializing every microstate.

7.4 The one real modeling decision: discretizing nucleosomes

TF occupancy is clean (binary per TetO, from `tfbs_{k}`). Nucleosomes are emitted by v5 as continuous-position segments and must be mapped to a discrete state variable:

- **Coarse (recommended to start): region indicators.** A few nucleosome “slots” (promoter nuc yes/no, one or two array-region slots) as binary features, coupled to TF/promoter state. Few parameters, trivial Z .
- **Fine: a proper lattice gas.** Bin the amplicon; a nucleosome occupies a run of about 147 bp of bins with exclusion; one energy per nucleosome. The positional degeneracy (a nuc can sit at many nearby positions) is integrated over automatically by the DP.

7.5 Minimal first model

1. **Features:** one TF field h (shared, or per-site), one nearest-neighbor cooperativity J , one TF-promoter-nucleosome antagonism term. About 3 parameters — close to the old `3param_nuc` / `3param_tfcoop` flavors, but cleanly derived.
2. **Empirical moments** from v5 calls: per-site occupancy, adjacent co-occupancy, TF/nuc co-occurrence.
3. **Z and model moments** via a transfer matrix over the array (+ a coarse promoter-nuc slot).
4. **Fit** by concave gradient ascent to moment-match; report energies with Hessian-derived CIs.

Two subtleties to revisit later: v5 calls are themselves inferred (could marginalize over v5’s posterior instead of hard calls if observation noise matters); and molecules with partial coverage should contribute to a feature’s moment only where they are informative.